Supermarket Sales Analysis

Developer Arena Internship • Project 1 • Retail Domain

Executive Summary

This project analyzes 2,000 supermarket transactions to identify sales patterns, best-performing product categories, customer behavior, branch performance, and peak sales periods. The analysis uses Python, Pandas, NumPy, Matplotlib, and Seaborn.

The dataset contains 2,000 records with 0 missing values and 0 duplicate rows. Total recorded sales were ■519,280.69, with an average transaction value of ■259.64.

Key Performance Indicators

KPI	Value
Transactions	2,000
Total Sales	■519,280.69
Average Transaction	■259.64
Missing Values	0
Duplicate Rows	0

Top Product Categories

Rank	Product Line	Sales
1	Health & Beauty	■96,970.32
2	Sports & Travel	■91,177.89
3	Electronic Accessories	■88,995.19
4	Home & Lifestyle	■86,351.63
5	Food & Beverages	■78,233.94
6	Fashion Accessories	■77,551.72

Health & Beauty generated the highest sales at ■96,970.32, while Fashion Accessories generated the lowest at ■77,551.72.

Branch Performance

Branch	Sales	Average Sale	Transactions
B	■180,821.79	■259.80	696
A	■179,047.93	■265.65	674
C	■159,410.97	■253.03	630

Sales Timing

Thursday recorded the highest total sales at ■79,450.41, while Sunday recorded the lowest at ■64,607.07. The peak sales hour was 18:00 with ■48,286.93.

Customer and Statistical Analysis

Member and Normal customer segments showed broadly similar sales performance. The mean total transaction value was approximately ■259.64 and the median was approximately ■258.04. The mean quantity was approximately 5.03 items per transaction, with a median of 5.

Correlation analysis showed very weak relationships among the main transaction variables, including Total Sales and Rating. This suggests that the observed sales values and customer ratings did not have a strong linear relationship in this dataset.

Business Insights

• Health & Beauty was the highest-selling product category.

• Branch B generated the highest total sales, while Branch C generated the lowest.

• Thursday was the strongest sales day and Sunday was the weakest.

• 6 PM was the peak sales hour, indicating an important evening demand period.

• Customer segments had relatively similar sales performance.

Recommendations

• Maintain sufficient inventory for high-performing categories such as Health & Beauty.

• Investigate lower-performing categories and branches for targeted promotions and operational improvements.

• Align staffing and inventory availability with peak evening demand, especially around 6 PM.

• Use targeted weekday and weekend promotions to address weaker sales periods.

• Continue monitoring customer ratings alongside sales to identify changes in customer experience.

Conclusion

The project demonstrates an end-to-end retail data analysis workflow including data validation, transformation, descriptive statistics, categorical analysis, time-based analysis, correlation analysis, visualization, and actionable recommendations. The resulting insights can support inventory planning, staffing, promotions, and branch-level decision making.